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Project Title:	Mechanotransduction Through Integrins		
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Task Description:	The specific aims of this project are 1. To compare the effects of mechanical stresses of different intensity, frequency and direction on focal adhesion assembly, cell strengthening and G _protein activation. 2. To analyze the Rho signaling mechanism responsible for cell strengthening behavior. 3. To identify the molecules that mediate transmembrane mechanochemical coupling between integrins and G proteins within the focal adhesion.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Elucidation of the mechanism by which integrins mediate cellular mechanotransduction should help us to understand the molecular basis of the cellular response to gravity as well as many other forms of mechanosensation and tissue regulation. Through the proposed research program, we hope to identify specific molecular mediators of mechanochemical transduction as well as novel mechanical signaling pathways in mammalian cells. These molecules may serve as potential targets for future modes of therapeutic intervention in diseases, such as atherosclerosis, hypertension, and osteoporosis, which are characterized by abnormal force transduction by cells. Further insight into mechanobiology also may impact on design of artificial tissues and medical devices for tissue repair and transplantation.
Task Progress:	Aim 1: The ability of cells to sense mechanical forces and respond by strengthening their integrin-extracellular matrix (ECM) adhesions is critical for cell viability as well as control of cell shape, growth and motility. To better understand how cells adapt to changes in mechanical forces exerted on their ECM adhesions, we applied tensional forces to magnetic microbeads bound to cell surface integrin receptors using magnetic pulling cytometry and measured dynamic changes in bead displacement with sub-micrometer resolution using optical microscopy. These studies revealed that cells exhibit four distinct types of mechanical responses: 1) an immediate viscoelastic response to a brief 3 sec force pulse that resulted in almost complete elastic recovery, 2) early adaptive behavior characterized by pulse-to-pulse attenuation in bead displacement in response to an oscillatory force regimen composed of multiple brief (3 sec) force pulses separated by short (4 sec) relaxation periods, 3) a later adaptive cell stiffening in response to sustained (> 15 sec) static stresses, and 4) a large-scale repositioning response in which beads exposed to prolonged (> 1 min) static stresses were retracted back by the cell against the applied force. Importantly, these forms of cellular adaptation differed biochemically. The immediate viscoelastic response and early adaptation behavior were prevented by dissipating cytoskeletal prestress (isometric tension) with inhibitors of either myosin ATPase or Rho-associated kinase (ROCK), whereas the later adaptive response to more prolonged (> 15 sec) stresses was not sensitive to either of these inhibitors. The large-scale repositioning of beads in response to sustained (> 1 min) stress was prevented by inhibiting myosin-based tension generation through interference with Rho signaling, similar to the case of the immediate and early responses, but in addition it was also prevented by inhibiting mechanosensitive ion channels using gadolinium. All of these responses were suppressed by cooling the cells to 48C to slow biochemical remodeling events, and the large-scale repositioning response required that integrins be chemically activated through occupancy of the RGD binding site. These results demonstrate that cells use multiple distinct mechanisms to sense and respond to static and dynamic changes in the physical properties of the ECM that alter the level of mechanical stress applied to cell surface integrin receptors. We have previously shown that integrins act as mechanoreceptors. Using magnetic twisting cytometry, we demonstrated that applying force to integrin receptors but not to metabolic receptors resulted in focal adhesion (FA) assembly, cell strengthening, cAMP production and gene expression. However, it remains unclear whether mechanical stresses chemically activate cell surface integrin receptors during the stress-induced FA assembly response. To determine whether force activates integrins, we stretched capillary endothelial cells cultured on flexible fibronectin-coated Bioflex substrates using a FlexCell system. Integrin activation was measured by staining with either antibodies that recognize active integrins (12G10 for beta-1 integrin) or fluorescent soluble fibrinogen that binds unoccupied active beta-3 integrins. Total cellular beta-1 integrins and phosphotyrosine also were visualized using M-106 antibodies and PY20 antibodies, respectively; FAK tyrosine phosphorylation was analyzed using the Western blotting technique. These studies revealed that when cells plated for 1 hr on the fibronectin substrates in chemically-defined medium were subjected to mechanical strain (15% equi-biaxial elongation) for 5 min, beta-1 integrins became activated, as indicated by their clustering and relocalization into FAs at the periphery, as detected using the 12G10 antibody. Preliminary studies suggest that mechanical force application also increase phosphotyrosine staining in FAs and FAK tyrosine phosphorylation. A similar level of mechanical strain applied for 15 min also increased beta 3 integrin activation in fibroblasts compared to non-stretched controls. These results suggest that mechanical stress results in integrin receptor activation as well as stimulation of down-stream signaling pathways. Aim 2: The small GTPase Rho plays a central role in integrin-dependent control of cell contractility, and has been shown to be activated by mechanical force application to surface integrin receptors. But the molecular basis of this mechanotransduction response remains unclear. To explore if mechanical force activates Rho downstream of integrin receptor activation, we plated endothelial cells on BioFlex membranes coated with a non-activating (K20) beta-1 integrin antibody and Rho activation was measured by either the Rhotekin pull-down assay (Western blotting) or a novel FRET-based assay (confocal microscopy) after mechanical stretching. Application of mechanical force (15% strain) for 15 min significantly increased Rho activation in these cells compared to control cells. To determine the possible mechanism of Rho activation, we measured the translocation of p190Rho-GAP from the cytosol to the cytoskeleton fraction because this is an upstream regulator of Rho activity. We found that mechanical stretch increased the translocation of both p190RhoGAP and the tyrosine kinase, c-src, to the cytoskeletal fraction, and that this response paralleled the time course of Rho activation. To further confirm that mechanical force application activated Rho, we transfected these endothelial cells with a RhoFRET probe (Raich-RBD) which detects the changes in endogenous Rho activity. FRET efficiency was measured using the acceptor-photobleach method. Preliminary results suggest that application of 15% mechanical strain to cells adherent to fibronectin-coated BioFlex plates for 15 min induced a statistically significant decrease in FRET efficiency which is indicative of Rho activation in response to mechanical force. Taken together, these results strongly suggest that mechanical force application through beta-1 integrins results in stress-dependent activation of Rho. Aim 3: In past work, we demonstrated that mechanical twisting of integrin receptors induces recruitment of the large G protein G-alpha s to FAs, a rise in cAMP and an increase in cAMP-dependent gene transcription. Recently, we further analyzed the role of heterotrimeric G proteins in this mechanotransduction response using melittin (an inhibitor of G-alpha s activity) and pertussis toxin (an inhibitor of G-alpha i) in a novel KID-KIX based assay that reconstitutes beta-lactamase activity through protein fragment complementation and thereby converts a green fluorescent dye (CCF2-AM) to a blue reporter signal within minutes after cAMP production is induced in the cell. NIH-3T3 fibroblast cells that stably express this KID-KIX reporter were cultured on type I collagen-coated Bioflex substrates and exposed to a cyclical mechanical load (15% strain, 20 cycles/min) for 5 min in the presence or absence of melittin (1uM) or pertussis toxin (200 ng/ml). Stress application significantly increased cAMP production in these cells compared to non-stretched control cells. Melittin treatment completely attenuated force-induced cAMP production confirming that G-alpha s mediates this process. In contrast, pertussis toxin increased cAMP production suggesting that G-alpha i may normally counterbalance force-induced cAMP production. To further confirm the role of G-alpha s in this mechanotransduction response, we measured the activation of CREB through nuclear staining of phosphoCREB. Preliminary results show that mechanical

force (15% cyclic strain for 5 min) increased the number of cells with phospho-CREB nuclear staining compared to non-stretched cells. Again, pre-treatment with melittin significantly inhibited the p-CREB nuclear staining. Collectively, these findings strongly suggest that G-alpha s is the key G-alpha protein that mediates force-induced cAMP production through integrin receptors.

We also carried out studies to identify the specific G proteins involved in this mechanotransduction response. cAMP production is generally due to activation of adenylate cyclase which is activated in a G -alpha s protein-dependent manner. Working with suspended capillary endothelial cells and 293T cells bound to 4.5 mm diameter magnetic beads coated with non-activating (K20 or MAB2000) or activating (BD15) beta-1 integrin antibodies, we found that clustering of integrins induced by binding to microbeads coated with both antibodies promoted recruitment of G-alpha s and G-beta proteins to FAs at the site of bead binding. Interestingly when the same experiments were performed on 293T cells over expressing wild type (Wt), dominant negative (DN) or constitutively active (CA) G-alpha s proteins, the G-alpha s proteins were equally recruited to bead-associated FA fraction regardless of their activity state. In contrast, significantly less G-beta proteins were present in this fraction in cells expressing CA-G-alpha s proteins. Similarly, we found that short isoforms of Wt, DN and CA G-alpha s proteins were detected at equal levels in immunoprecipitates obtained using anti-beta-1 integrin antibodies, even though the levels of G-beta proteins level were increased in the Wt- and DN-Gas samples. Past work by others has shown that although DN-Gas proteins are always detected as a heterotrimeric complex (due to the association of the alpha, beta, and gamma subunits), CA-G alpha s proteins do not always associate with G-beta proteins. Our results raise the possibility that the association of heterotrimeric G proteins with beta1 integrin is mediated via the G alpha s subunit, independently of the G-beta-gamma complex.

We also investigated the effect of mechanical stress applied to beta-1integrins using magnetic twisting cytometry on the level of recruitment and activation of G-alpha s. These studies revealed that twisting of clustered but non-activated beta-1 integrins increased the recruitment of G-alpha s in the bead fraction after 3 min of mechanical stimulation, although the amount present decreased at later times. In contrast, significant activation of G-alpha s was observed only after 10 min of mechanical twisting of beta-1 integrins that were both clustered and not biochemically activated, possibly because force application to integrins bound to activating antibodies already bound optimal levels of G-alpha s, at least under the conditions we carried out this study. Western blot analysis of bead-associated FA fractions carried out in parallel, and prepared from cells loaded with biotinylated-azidoanilido-GTP (AA-GTP), also showed that only the short isoform of G-alpha s (43 Kda) was activated after mechanical stress was applied to activated b1 integrin as indicated by staining with HRP-streptavidin.

Bibliography Type:	Description: (Last Updated: 02/25/2021)
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Awards	Ingber DE. "Pomerat Memorial Lecturer, Cell Biology Graduate Program, University of Texas, November 2004." Nov-2004
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Presentation	Ingber, D.E. "Cell Engineering and Tissue Morphogenesis " Current Progress in Tissue Engineering Meeting, Cambridge, MA Oct-2004
Presentation	Ingber, D.E. "The Living Cell as an Integrated Mechanochemical System" Single Cell Mechanics Conference, MIT, Cambridge, MA Oct-2004
Presentation	Ingber, D.E. "Biological Design Principles for the Nanotechnologist" New York Academy of Science Frontiers in Science Meeting, NYC Mar-2005
Presentation	Ingber, D.E. "Mechanical Control of Tissue Development" Annual Meeting of the American Association for the Advancement of Science, Washington, D.C Feb-2005
Presentation	Ingber, D.E. "Structural determinants of biological control: from nanomachines to living cells and tissues. " American Physical Society Annual Meeting, Los Angeles, CA Mar-2005
Presentation	Ingber, D.E. "Cellular Engineering and Tumor Angiogenesis" Williams College Premedical Society, Williams, MA Apr-2005

Presentation	Ingber, D.E. "Cellular Tensegrity and Mechanotransduction " Talbot Memorial Lecture in Theoretical Mechanics, University of Illinois, Urbana Champaign, Champaign, IL Mar-2005
Presentation	Ingber, D.E. "Cellular Tensegrity, Mechanotransduction and Tissue Morphogenesis " Pomerat Memorial Lecture, Cell Biology Graduate Program, University of Texas, Galveston, TX Nov-2004